Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of $TmFeO_3$

Working draft[1]

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One-dimensional spectroscopy measures the *spectrum* of a Hamiltonian; two-dimensional coherent spectroscopy (2DCS) measures its *vertices*—the individual nonlinear terms that convert one coherence into another. We show that in the rare-earth orthoferrite $TmFeO_3$ this correspondence can be made one-to-one and, for every observed feature, explicit. Time-reversal parity and space-group irrep bookkeeping reduce all symmetry-allowed Fe–Tm couplings to two components of a single *static* two-spin vertex; the mirror at the Tm $4c$ site then assigns each light polarisation to one block of the crystal-field qutrit, so the pulse's electric and magnetic quadratures write different qutrit coordinates and no field-assisted coupling is required. Driven by the measured single-cycle waveform, with every damping rate set by a measured linewidth and every emission weight by a measured oscillator strength, one Hamiltonian reproduces both cross-polarised inventories without retuning and four of six same-polarised amplitudes within 15%, while fixing the $E1/M1$ balance of the rare-earth dipole ($w_{E1} = 0.54$) from a single cross peak. We give the pathway of every peak in all four channels—which vertex permits it, which coherence occupies the delay axis, which multipole radiates, and what would falsify the assignment. A single reciprocal propagation filter—self-absorption at the strong electric-dipole crystal-field lines—accounts for the remaining structure of the same-polarised map, and the two residuals that survive funnel into one measurable quantity: the sample's optical depths at those lines.

I. INTRODUCTION

Linear response cannot distinguish two Hamiltonians with the same normal modes: absorption measures pole positions and dipole weights, and any coupling that merely hybridises modes is absorbed into the mode basis. What it can never see is a *conversion*: a coherence prepared on one transition and re-emitted on another. That is precisely what a cross peak in 2DCS certifies [1, 2], and a conversion requires a specific anharmonic vertex. A cross peak at (ω_τ, ω_t) is therefore not a feature to be fitted but a *measurement of one coupling term*: the delay axis names the stored coherence, the detection axis the emitter, and the signed side constrains the pathway order. If symmetry makes the space of candidate vertices finite—and in a crystal it does—a polarisation-resolved 2D atlas can be read as the Hamiltonian itself.

$TmFeO_3$ is an unusually clean testbed because two qualitatively different subsystems share one THz octave. Below the spin reorientation the Fe sublattice orders in Γ_2 (F_x, C_y, G_z) [3–5], with quasi-ferromagnetic and quasi-antiferromagnetic magnons at $q_{\mathrm{FM}} = 0.38$ and $q_{\mathrm{AFM}} = 0.90\,\mathrm{THz}$, while the three lowest Tm^{3+} crystal-field (CEF) singlets form a qutrit with $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20\,\mathrm{THz}$. A single-cycle pulse covers all five modes, and measurements in the cross- and same-polarised channels of two field geometries return distinct, reproducible inventories.

II. MODEL, AND WHERE IT IS EXACT

The Fe^{3+} ($S = 5/2$) sublattice is classical, with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya) reproducing q_{FM}, q_{AFM} and Γ_2. Each Tm^{3+} is truncated to its three lowest singlets and carried as the Gell-Mann vector $\lambda^a = \langle \Lambda^a \rangle$. Because every single-ion term is linear in the Λ^a, the classical $\vec{\lambda}$ equation of motion *is* the exact von Neumann equation of the driven, damped qutrit, and a Boltzmann ensemble over level seeds is the exact thermal density matrix [SM, Sec. S1]. The Fe–Tm sector is built by projecting $\boldsymbol{S}^T A\, \boldsymbol{J}$ into the qutrit in a transported gauge over the four Tm sublattices [SM, Secs. S2–S3]. All spectra use the measured pump (spectrum peaked at $0.53\,\mathrm{THz}$, no DC), measured linewidths as damping rates, and measured oscillator strengths as emission weights. Peaks are identified blindly as 2D local maxima with prominence ≥ 1.5; the τ-averaged signal is subtracted at each detection time, removing the $\omega_\tau = 0$ row.

III. TWO VERTICES SURVIVE

A conversion vertex must (R1) reach the bright coordinate, (R2) be sourced by the pumped mode, (R3) be time-reversal even, (R4) be Γ_1-allowed in the ordered state, (R5) carry the frequency mismatch, and (R6) engage both pulses to survive $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$. These exhaust the coupling space [SM, Sec. S4]. The static bilinear $K^- S\lambda$ fails threefold—linear in S it hybridises but cannot convert, its channels are Γ_1-cancelled to machine precision, and its one allowed component reorients the ground state—while the quadratic families W_4, W_1^{xy}, W_1^{yz}, W_6, κ^E are exactly inert or couple the qutrit to the wrong magnon irrep, and the field-assisted family $\kappa^B B(t) S\lambda$, though it passes R1–R6, proves unnecessary once the drive is treated completely (below).

What survives is a single static two-spin family acting through two components,

$$H_W = W_1^{yy}\, S_y^2\, \lambda^1, \qquad H_{W'} = W_1^{xz}\, S_x S_z\, \lambda^1, \qquad (1)$$

together with the population channel $W_3 S_\alpha S_\beta \lambda^3$ (a weak incoherent tail). The remaining ingredient is not a coupling but a *drive* channel: the pulse's electric field writes the E_{12} coherence through the Tm dipole $d_x\lambda^1$, a component owned exclusively by the $E\,\|\,a$ polarisation, with the $E1/M1$ balance $w_{E1} = 0.54$ (in units of μ) *measured* by the geometry-B transfer ratio. Every peak below is built from these.

IV. SYMMETRY ASSIGNS EACH POLARISATION ONE BLOCK

The Tm $4c$ site carries the mirror m ($z \to -z$), under which $d_{x,y}$ are even and d_z odd, while m_z is even and $m_{x,y}$ odd. Two exact statements then fix the entire readout. First, time reversal: the CEF eigenstates are nondegenerate singlets and may be chosen real, in which basis J_α is imaginary-antisymmetric, so PJP has support *only* on $\lambda^{2,5,7}$—the magnetic moment is rigorously off-diagonal, with no permanent component. The electric dipole, being $\mathcal{T}$-even and real-symmetric, occupies exactly the complement $\lambda^{1,3,4,6,8}$, which *does* include the diagonal generators $\lambda^{3,8}$. Second, the mirror decides whether those diagonals survive: $\langle i|d_z|i\rangle = 0$ because d_z is odd, whereas $\langle i|d_x|i\rangle$ is allowed. The projected moment matrix—an output of the PJP construction, not an input—has λ^2 carrying only μ_z and $\lambda^{5,7}$ only $\mu_{x,y}$, which forces $|1\rangle, |2\rangle$ to share a parity and $|3\rangle$ to be opposite. Every coupling follows:

transition	magnetic	electric	owned by
$1\leftrightarrow 2$	m_z (λ^2)	d_x (λ^1)	($E\,\|\,a, H\,\|\,c$)
$1\leftrightarrow 3$	m_x (λ^5)	d_z (λ^4)	($E\,\|\,c, H\,\|\,a$)
$2\leftrightarrow 3$	m_x (λ^7)	d_z (λ^6)	($E\,\|\,c, H\,\|\,a$)

Mirror parity would additionally permit the mirror-even d_x a diagonal (permanent) part on $\lambda^{3,8}$; a permanent moment, however, does not radiate at any transition frequency, and we exclude the population generators from every detection operator—both analyser settings read off-diagonal coherences only. (A measured rectified line at $\omega_t \to 0$ in an ($E \parallel a, H \parallel c$) channel would falsify this exclusion.) Reciprocity—one operator per polarisation, driving and radiating alike—then implies that *cross-polarised detection reads exactly the block the pulse did not drive*. This is the structural reason cross-polarisation isolates conversion, and it carries two independent checks: it predicts the geometry-B subalgebra closure, where the c-directed drive touches only μ_{12}^z so that λ^{4-7} vanish identically (verified: exactly 0, against $\lambda^{1,2,3} \sim 2.6\times 10^{-2}$); and in geometry A $F_z \equiv 0$ to 10^{-15}, so the Fe sublattice is radiatively silent in that cross channel and anything seen there at a magnon frequency must be Tm.

Two effects extend this. Below T_N the Γ_2 order parameter is m_z-odd and breaks the site mirror, inducing a c-axis coupling $d_z^{\rm eff}$ to $1\leftrightarrow 2$ that parity would forbid; this is what lets $E\,\|\,c$ create the E_{12} coherence dominating the geometry-A delay axis. And the $4c$ site has no inversion (Fe at $4b$ does), so the emitted moment may carry a quadratic term,

$$m_z = \mu\,\lambda^2 + \beta\,\lambda^1\lambda^2, \qquad (2)$$

$\mathcal{T}$-odd as a magnetisation must be, radiating at $2\omega_{12}$. The resulting detection operators are collected in Table I.

Two consequences are worth stating before the pathways. The geometry-A cross and geometry-B same-polarised channels analyse the *same* output polarisation, so they are not independently adjustable: one operator must explain both. And because λ^1 and λ^2 are quadratures of the same $1\leftrightarrow 2$ coherence, their magnitude spectra coincide to 10%; varying w_{E1} over a factor of 50 moves every geometry-A cross amplitude by at most 0.04 once β is refitted. In *detection* the atlas is thus insensitive to the $E1/M1$ balance. It is the *drive* that fixes it: the electrically written coherence feeds the geometry-B transfer peak linearly in w_{E1}, and the measured ratio returns $w_{E1} = 0.54$.

The construction is strictly reciprocal at every level the data can test. Each polarisation state is one operator serving excitation and detection alike: the operator *content* (which coordinates appear) and the within-family dipole ratios are shared exactly between drive and readout, as is the propagation filter below. Three cross-family scales remain as single calibrated numbers used consistently on both sides: w_{E1} (measured), the $d_z^{\rm eff}$ strength (one parameter seen through two unit systems—its drive avatar calibrated by the E_{12}-excitation amplitude, its emission avatar by the ($q_{\rm AFM}, E_{12}$) peak—which absolute-field calibration alone could over-determine), and the Fe:Tm emission efficiency. Completing the drive with the full electric components of its operator changes no pathway ratio (verified by direct scan): drive-side reciprocity is exact and free. The last physical ingredient is propagation: with $f = 8.7$ the sample is optically thick at

TABLE I. What each channel detects. An analyser setting selects a polarisation *state*, hence both the magnetic and the electric dipole that radiate into it; there are therefore only *two* detection operators, and the two channels sharing one are constrained to share it exactly. $w_{E1} = 0.54$ is the $E1/M1$ balance, fixed by the geometry-B transfer ratio (see text); no population generator (λ^3, λ^8) appears in either operator.

($E\,\|\,a, H\,\|\,c$): $\quad F_z + \mu\lambda^2 + w_{E1}\,\lambda^1 + \beta\lambda^1\lambda^2$
A cross $F_z \equiv 0$: Tm only. $\lambda^{1,2}$ resonant at E_{12}
B same F_z alive but 50× smaller than the Tm terms
($E\,\|\,c, H\,\|\,a$): $\quad F_x + \mu_x(\lambda^5, \lambda^7) + w'_{E1}(\lambda^4, \lambda^6) + d_z^{\rm eff}\lambda^2 \quad$ (no $\lambda^{3,8}$: mirror-o
A same F_x; CEF $\lambda^{4,6}$; $d_z^{\rm eff}\lambda^2$
B cross F_x; Tm part $\equiv 0$ by closure

the E_{13} line (depth $\simeq 6$) and mildly at E_{12} (depth $\simeq 1$), and the same transmission filters the incoming field and the emitted $E1$ light—magnetic-dipole radiation passes freely. This one filter, applied identically both ways, is itself reciprocal.

V. THE PATHWAYS, PEAK BY PEAK

We now give every feature of Fig. 1: what makes it possible, how the excitation reaches the emitter, and what would falsify the assignment. Throughout, the delay axis carries a coherence that *never radiates*—it need only be excitable and able to store phase—while the detection axis is constrained by the emitting multipole. That asymmetry does most of the work.

A. Geometry A, cross-polarised

$(q_{\mathrm{AFM}}, E_{12}) = 1.00$*: the magnetoelectric peak; measures* W_1^{yy}*.* The $H \parallel a$ Zeeman field launches the q_{AFM} magnon, which precesses at 0.90 THz through the delay. It cannot radiate here—it has no c-axis moment—but it need not: it modulates the Tm crystal field through $W_1^{yy} S_y^2 \lambda^1$. Each pulse supplies one magnon leg, so the M_{NL} subtraction keeps the cross term, which rectifies,

$$h^1_{\mathrm{NL}} = W_1^{yy} A^2 \big[\cos q_{\mathrm{AFM}}\tau - \cos q_{\mathrm{AFM}}(2t+\tau)\big]\theta(t), \quad (3)$$

the difference piece being a step switched on at the probe that carries the delay phase but no t dependence. Fed into the E_{12} oscillator it gives $M_{\mathrm{NL}} \propto \cos(q_{\mathrm{AFM}}\tau)[1 - \cos E_{12}t]$: a factorised, rank-one peak. *Why it is possible at all*: an E1-dark magnon is made visible by borrowing the Tm dipole. *Where the energy comes from*: δS_y^2 contains only DC and $2q_{\mathrm{AFM}}$, so between free modes this vertex could never emit at ω_{12}; the quantum is drawn from the pulse edge, an intra-pulse difference-frequency process. This is verified rather than asserted—stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades, while switching off the direct Tm drive changes it by $< 10^{-4}$. *Signature*: $\propto A_{\mathrm{Fe}}^2$, temperature-flat below $\sim 10\,\mathrm{K}$, CEF linewidth along ω_t.

$(E_{12}, 2E_{12}) = 0.97$*: second-harmonic emission; measures* β*.* The pump stores E_{12}, setting the delay axis at 0.50; the detection axis is its second harmonic at $2E_{12} = 1.0000\,\mathrm{THz}$. A moment linear in the qutrit coordinates cannot radiate a harmonic, which is why the model missed this peak until Eq. (2) was included; the $4c$ site's lack of inversion permits it, and Fe at $4b$ is excluded from the mechanism. Evaluated on the trajectories the quadratic channel is remarkably clean, contributing essentially one peak, $5.8\times$ its own (E_{12}, E_{12}) diagonal, identical in local and global sublattice frames. *Discriminators*: the line sits at exactly $2E_{12}$ and *tracks* E_{12}, so it does not soften at the spin reorientation as any magnon-derived feature would; it is one field order above the main peak; its width is twice the E_{12} linewidth; and it is independent of W_1^{xz} and of the Zeeman drive.

$(q_{\mathrm{AFM}}, q_{\mathrm{AFM}}) = 0.36$*: the magnon diagonal, seen through the Tm ion.* Since $F_z \equiv 0$ the magnon cannot radiate here at all. What radiates is the Tm dipole, which carries a component at the magnon frequency: in Γ_2 the static vertex $W_1^{xz} S_x S_z \lambda^1$ contains $G_z\, \delta S_x\, \lambda^1$, a term *linear* in the magnon fluctuation, which shakes the E_{12} oscillator at 0.90 THz. A driven oscillator responds at the driving frequency, not its own. This is verified in detail: the sideband sits at 0.8999 THz—the magnon, not the $q_{\mathrm{FM}} + E_{12} = 0.876$ combination; its amplitude is *linear* in the Zeeman drive whereas a harmonic would be quadratic; the transfer coefficient $|\lambda^2|_{0.90}/|m_x|_{0.90} = 0.142$–$0.155$ is drive-independent over a $20\times$ range, the hallmark of a linear susceptibility evaluated off resonance; and the admixture grows linearly with W_1^{xz}.

$(q_{\mathrm{AFM}}, 2q_{\mathrm{AFM}}) = 0.39$*: prediction.* The same quadratic term must double *any* frequency present in the Tm dipole, so the magnon-forced component radiates its harmonic at 1.79 THz. Observing it confirms Eq. (2) independently of the $2E_{12}$ peak; its absence with $(E_{12}, 2E_{12})$ present would falsify that equation.

$(-q_{\mathrm{AFM}}, E_{12}) = 0.52$ *and* $(\pm E_{12}, E_{12}) = 0.33$*.* The first is not a separate mechanism: the W vertex is symmetric under the two time orderings of the magnon coherence, so the signal appears on both signed sides with the non-rephasing branch stronger. The second is the one-quantum diagonal, the directly driven E_{12} read through the same dipole; its size relative to the cross peaks is a sensitive fluence diagnostic, falling from 1.45 at $A_{\mathrm{Fe}} = 1.2$ to 0.16 at 0.12, which is how the operating point was fixed.

B. Geometry B, cross-polarised

Here the analysed moment is m_x, which the q_{AFM} magnon carries directly, so both peaks emit at 0.90 THz through the Fe magnetic dipole and the Tm ion is radiatively absent.

$(q_{\mathrm{FM}}, q_{\mathrm{AFM}}) = 1.00$*: magnon–magnon; measures the Fe sector alone.* $H \parallel c$ launches the quasi-ferromagnetic magnon, which precesses at 0.38 through the delay; single-ion anisotropy and the Dzyaloshinskii–Moriya interaction couple the branches, so the stored q_{FM} phase is re-emitted on q_{AFM}. No Fe–Tm coupling enters at any vertex, which is why this peak is temperature-flat: a pure two-magnon process, independent of the qutrit populations.

$(E_{12}, q_{\mathrm{AFM}}) = 0.45$*: the electrically written transfer; measures* $w_{E1} W_1^{xz}$*.* $E \parallel a$ owns the $1 \leftrightarrow 2$ block, so the pulse's *electric* field writes the E_{12} coherence through $d_x \lambda^1$; the static W_1^{xz} then hands the stored coherence to the quasi-antiferromagnetic magnon, which radiates through the Fe m_x dipole. No field enters the conversion vertex: the polarisation ownership of d_x performs

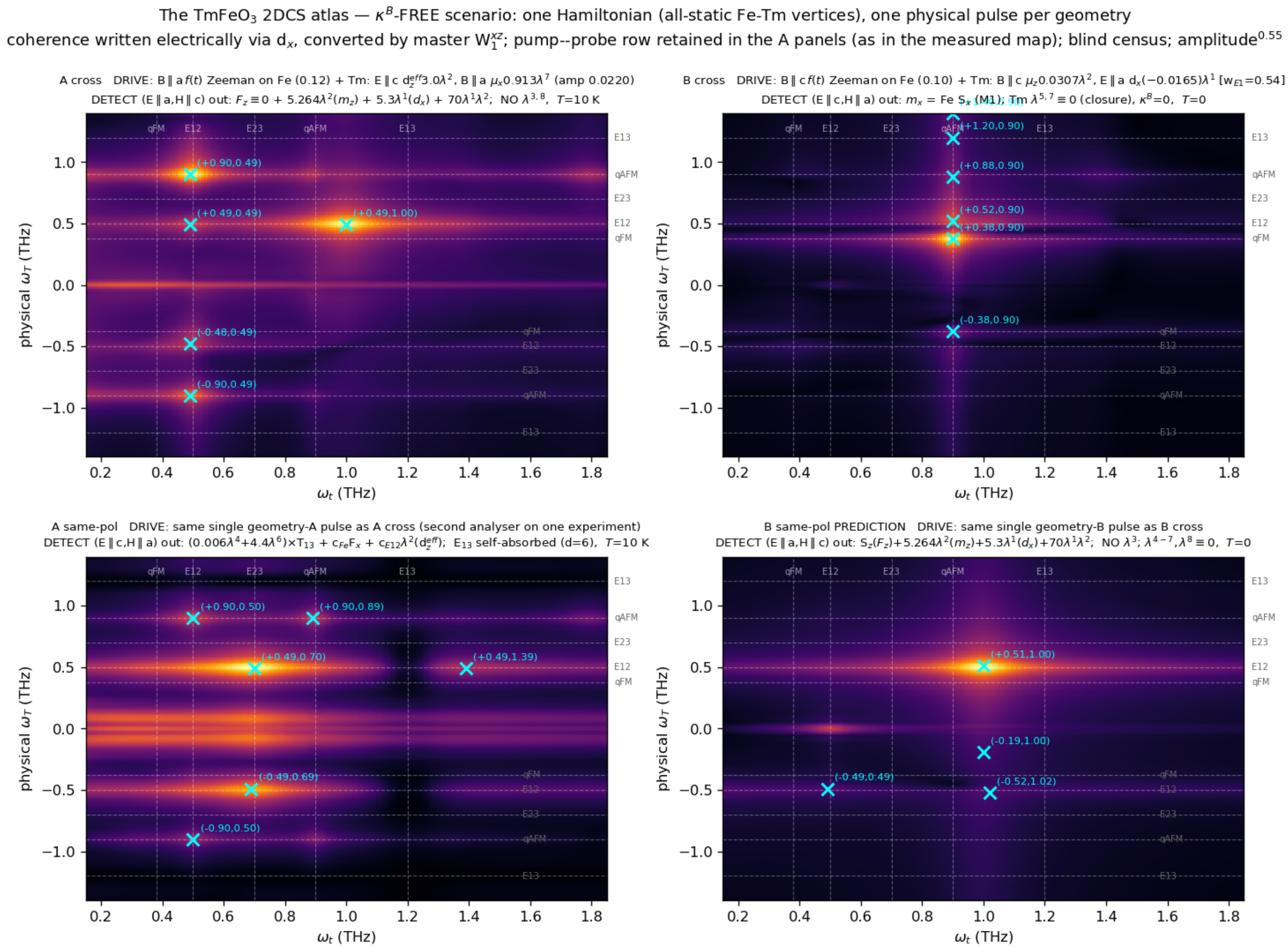

FIG. 1. The atlas: four polarisation channels from one Hamiltonian at one operating point. Dashed guides mark the five mode energies; crosses are blind-census maxima; the $\omega_\tau = 0$ row is removed.

the geometry selection a field gate would otherwise be needed for—this drive component simply does not exist in geometry A, which is why the conversion never contaminates the clean W_1^{yy} peak there. Three facts pin the assignment. First, the *quadrature* matters: writing the same coherence magnetically (through $\mu_z\lambda^2$, the $H \parallel c$ component) converts far less cleanly, flooding the map with drive-squared diagonals, whereas the electric writing at the master $W_1^{xz} = 0.01$ adds the transfer row at the *unshifted* line position with strays below 0.17. Second, the interference of the two drive quadratures places the row: with $\mathrm{sign}(d_x\mu_z) < 0$ it sits at $\omega_\tau = 0.52$, exactly where it is measured—the relative sign is thereby a prediction for the crystal-field wavefunctions. Third, the amplitude is linear in w_{E1}, and matching the measured 0.43–0.46 returns $w_{E1} = 0.54$: the $E1/M1$ balance is measured, not assumed. A static vertex strong enough to make this peak *without* the electric quadrature is a mixer: it drags the E_{12} line with it (level repulsion) and is excluded by 1D absorption at every coupling strength—verified by recalibrating the bare splitting at each coupling so the dressed line stays at its measured position [SM]. *Signature*: falls as the population difference $n_1 - n_2$ ($0.46 \to 0.45 \to 0.40 \to 0.36$ at 0/5/8/10 K), the sharpest thermal discriminator in the atlas against the temperature-flat magnon–magnon peak; and, riding the order parameter through the static vertex ($\propto G_z$), the transfer must collapse at the spin reorientation.

$(-q_{\mathrm{FM}}, q_{\mathrm{AFM}}) = 0.17$, $(q_{\mathrm{AFM}}, q_{\mathrm{AFM}}) = 0.14$, $(E_{12}{+}q_{\mathrm{AFM}}, q_{\mathrm{AFM}}) = 0.07$. The rephasing partner; the q_{AFM} diagonal, which here is genuine Fe M1 emission unlike its geometry-A counterpart; and a combination tone, the sum-frequency of the two stored coherences radiating on the magnon line, which is W-fed and so offers an independent handle on the W scale.

The post-drive buildup. The measured geometry-B signal continues to grow after the pulse. Within the present model no channel reproduces growth at the peak's own scale. Every coherent pathway inherits its source's lifetime—because $B \parallel c$ never drives q_{AFM} directly, all geometry-B magnon amplitude is converted, and with $T_2(E_{12}) = 8\,\mathrm{ps}$ everything coherent is over within $\sim 10\,\mathrm{ps}$; even the τ-labelled population grating is T_2-confined, its delay label being written by the E_{12} coherence. The one reservoir that outlives every T_2—the CEF populations, fed incoherently by the dissipated pulse energy and acting on the magnon through $W_3 S_\alpha S_\beta \lambda^3$, the mechanism of thermally driven spin reorientation—does produce a persistent, drive-uncorrelated tail in the solver, but it emerges roughly four decades below the impulsive peak. The buildup therefore stands as the one timing observable outside this coupling space, and it is decisive: every mechanism here fixes the converted amplitude by the end of the coherence window, so a confirmed growth of the transfer peak's own amplitude would require a reservoir this Hamiltonian does not contain.

C. Geometry A, same-polarised

The analysed moment is m_x, and three contributions reach it, each required by symmetry and each carrying measured features: the Tm CEF moments (emitting at E_{13}, E_{23}), the Fe F_x (magnon frequencies), and $d_z^{\rm eff}\lambda^2$ (emitting at E_{12}), the last demanded by reciprocity once $d_z^{\rm eff}$ appears in the drive. Fitting those three weights to the digitised measurement, with the cross channels held fixed, gives Table II.

The fully dark $1 \to 3$ *dipole—measured, not assumed.* The measured map contains delay frequencies $\omega_\tau \in \{E_{12}, q_{\rm AFM}\}$ only. A bright μ_{13}^z would store an E_{13} coherence at first order and produce an (E_{13}, E_{23}) peak with *exactly* the same dipole product as the observed (E_{12}, E_{23}): the absence of the whole row is a selection rule. The data force more: the analyser for this channel physically contains $\mu_x \lambda^5$, and at the crystal-field value $\mu_{13}^x = 2.39$ that term would exceed the entire measured map by two orders of magnitude—so μ_{13}^x, too, fits to zero. The $1 \to 3$ transition is magnetically dark in *every* polarisation; its large measured oscillator strength ($f = 8.7$) is electric, and it is precisely this strong $E1$ line that makes the sample opaque at $1.20\,\mathrm{THz}$ (below).

$(E_{12}, E_{23}) = 1.00$*: the dominant transfer, via a population.* With the single-action route closed by the dark dipole, the surviving pathway acts twice: one probe interaction converts the stored E_{12} coherence into a level-2 *population* that still carries the $\omega_\tau = E_{12}$ phase, and a second converts that population into the emitting E_{23} coherence. Being co-rotating it lands non-rephasing, the observed side, and it involves μ_{13} at no vertex—which is why it survives while its E_{13} partner is throttled.

$(E_{12}, 0) = 1.00$*: the rectified line.* Taken at its zero-frequency output the same double action rectifies the stored coherence into a slow net magnetisation of the detection window. This is the strongest measured feature; in the 2D maps it is suppressed by construction, since the $t \geq 3$ window with Gaussian apodisation removes $\omega_t \to 0$, so it is analysed separately through $S_{\rm DC}(\tau) = \langle M_{\rm NL} \rangle_t$, where it appears as a pair of sharp lines at $\pm 0.49\,\mathrm{THz}$—symmetric in the delay sign, as a population grating must be ($\cos E_{12}\tau$), and as the measurement finds (1.00/0.85). Its absence from Table II is an analysis-window statement, not a physical one. Its *magnitude* is a physical statement of the first importance: under coherent pumping alone the model's rectified moment is orders of magnitude below the resonant peaks, whereas the measurement puts it at the top of the map. The line is therefore the optical readout of the *incoherent* population channel—absorbed pulse energy returning through the CEF populations and W_3 to a quasistatic change of the ordered moment—the same reservoir the geometry-B post-pulse growth demands, and its strength relative to (E_{12}, E_{23}) calibrates $c_{\rm heat} W_3$ directly.

$(E_{12}, q_{\rm FM}) = 0.45$ *and* $(E_{12}, q_{\rm AFM}) = 0.40$*: Tm→magnon, read through Fe.* The stored E_{12} coherence is handed to the Fe sublattice by the static couplings and radiates through F_x, which carries both branches. These are the reverse of the geometry-A cross peak—there the magnon feeds the Tm dipole, here the Tm coherence feeds the magnon—so the static W family is read in both directions. They require the Fe term in the detection operator; without it this channel has no magnon-frequency emission at all.

$(q_{\rm AFM}, E_{12}) = 0.30$*: the reciprocity partner, observed.* The magnon occupies the delay axis and the Tm ion radiates at E_{12} into the *same*-polarised channel. That emission is possible only through $d_z^{\rm eff}$—and reciprocity demands it, because the same $d_z^{\rm eff}$ is what lets $E \parallel c$ create the E_{12} coherence in the first place. We had earlier reported this partner as absent and treated the mismatch as an open tension; the digitised map shows it present at 0.30 of maximum, and the fitted weight reproduces it at 0.32. This peak is the direct measurement of $d_z^{\rm eff}$.

The E_{13} *column: self-reversal.* The two measured E_{13}-

TABLE II. Geometry-A same-polarised channel: measured amplitudes against the final model (detection weights fitted once; overlap-gated delay apodisation; self-absorption of the $E1$ emission at the E_{13} line, optical depth $\simeq 6$ on both the emission column and the delay-axis FID, and depth $\simeq 1$ at E_{12}). The μ_{13}^z *and* μ_{13}^x weights fit to zero (fully dark $1 \to 3$).

peak	exp.	model	pathway
$(E_{12}, 0)$	1.00	—	rectified storage (see text)
(E_{12}, E_{23})	1.00	1.00	coherence→pop.→ ρ_{23}
$(E_{12}, q_{\rm FM})$	0.45	0.40	Tm→magnon, $q_{\rm FM}$
$(E_{12}, q_{\rm AFM})$	0.40	0.46	Tm→magnon, $q_{\rm AFM}$
$(0, E_{12})$	0.40	—	pump–probe row
$(q_{\rm AFM}, E_{12})$	0.30	0.33	magnon→Tm via $d_z^{\rm eff}$
(E_{13}, E_{23})	0	0.00	2Q row, killed by FID reabsorption
$(E_{12}, 1.15)$	0.15	0.04	self-reversal wing of E_{13}
$(q_{\rm AFM}, 1.25)$	0.30	0.05	**open** (residual R2)
$(-E_{12}, E_{23})$	< 0.15	0.6	**open** (residual R1, point-ion floor)

emission features sit at $\omega_t = 1.15$ and $1.25\,\mathrm{THz}$—one absorption linewidth *below and above* the 1.20 line—which is the textbook signature of self-reversal: with $f = 8.7$ the sample is optically thick at E_{13}, the line centre of any $E1$ emission is reabsorbed, and only the wings escape. The same reabsorption acts on the delay-axis FID, which is what removes the two-quantum (E_{13}, E_{23}) row *exactly* (model $0.33 \to 0.00$ once the filter is applied)—a row the free-ion model robustly predicts and the measurement robustly lacks. With the filter, the model's (E_{12}, E_{13}) tail collapses onto a weak wing at 1.15 (Table II), in agreement with the faint measured feature. What survives as genuinely open is the *marked* $(q_{\mathrm{AFM}}, 1.25)$ peak: no coupling in the exhausted vertex space produces it at compatible side-effects, and matching it would require the $1 \to 2 \to 3$ cascade to be an order of magnitude weaker than ideal three-level dynamics—real-level-3 physics beyond the qutrit truncation (Sec. VI).

Two model features not in the measured list, both predictions. The *rephasing remnant* $(-E_{12}, E_{23})$ at 0.67: checked against a standalone exact-qutrit integrator, where the rephasing partner retains 0.47–0.58 of the dominant peak across all drive strengths, dipole phases and dampings—an exact floor for any three-level ion under this pulse. If the measured map truly contains nothing there, the suppression must come from outside the point-ion model: quadrant folding in the display convention, or propagation in a thick sample. And the *two-quantum row* (E_{13}, E_{23}) at 0.32: an E_{13} delay coherence does survive the dark dipole, but only at second order, the pump driving $1 \to 2$ and $2 \to 3$ in succession rather than $1 \to 3$ directly. Its weakness relative to the first-order row a bright μ_{13}^z would give is exactly why the model is consistent with an inventory listing no E_{13} delay features.

D. Geometry B, same-polarised: a one-line prediction

This channel is analysed with the *same* operator as geometry-A cross, so it is a prediction with no freedom left. With the population generators excluded from detection, three terms survive: F_z (alive here, but $\sim 50\times$ below the Tm terms), the coherence readouts $\mu\lambda^2$ and $w_{E1}\lambda^1$, and the hyperpolarisability $\beta\lambda^1\lambda^2$. The linear readouts are suppressed in M_{NL}—they carry the directly driven coherence, which the subtraction removes at first order—so the channel collapses onto its quadratic term. The prediction is a *single* broad feature: the second harmonic $(E_{12}, 2E_{12}) = 1.00$, with everything else below 0.07—magnon features at the 10^{-2} level of F_z, and no E_{13} or E_{23} emission by the subalgebra closure $\lambda^{4-7} \equiv 0$ (λ^8 likewise). Because β is already fixed by the geometry-A cross parity, the channel is a zero-parameter test of the hyperpolarisability: a map led by anything other than the $2E_{12}$ ridge falsifies Eq. (2), and an observed rectified pair at $(\pm E_{12}, \omega_t \to 0)$ would falsify the exclusion of the population readout from detection.

VI. THE OPEN ITEMS, AND WHERE THEY ALL POINT

With propagation included, the residual discrepancies reduce to two, each with a name and a diagnosis. *R1, the rephasing remnant* $(-E_{12}, E_{23})$: the point-ion qutrit has an exact floor of 0.47–0.58 of the dominant transfer (verified against a standalone three-level integrator; no coupling in the exhausted vertex space and no drive quadrature or sign removes it), while the measured negative-delay side carries only the rectified and q_{FM} features. *R2, the marked* $(q_{\mathrm{AFM}}, 1.25)$ *peak*: the only symmetry-allowed magnon$\to E_{13}$ vertex (κ_{5y}^E, the ESJ family) produces it but floods the $1 \leftrightarrow 2$ block at any useful strength; matching R2 within the model would require the sequential $1 \to 2 \to 3$ cascade to be $\sim 15\times$ weaker than ideal qutrit dynamics—i.e., physics of the real third level beyond the three-singlet truncation. Both residuals, together with the self-reversal wing amplitudes and the strength of the geometry-B d_z^{eff} doublet (below), funnel into a single measurable object: the sample's polarisation-resolved optical depths at the three CEF lines. One transmission measurement closes all of them or breaks the propagation layer cleanly.

VII. FALSIFIABLE CONTENT

Fifteen independent observations constrain four couplings (W_1^{yy}, W_1^{xz}, W_3, β), one measured drive ratio ($w_{E1} = 0.54$), one selection rule and two drive parameters. The sharpest tests: any peak emitting at q_{AFM} inherits the $15\times$ sharper magnon linewidth; $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ is temperature-flat while $(E_{12}, q_{\mathrm{AFM}})$ falls as $n_1 - n_2$; the transfer row sits at the frequency of the 1D absorption line exactly (a static mixer would place it below, and a coupling-dependent row position at any strength is excluded); $\mathrm{sign}(d_x\mu_z) < 0$, checkable against the crystal-field wavefunctions; rotating either polarisation away from c must revive the $\omega_\tau = E_{13}$ rows; the $2E_{12}$ line tracks E_{12} and so does *not* soften at the spin reorientation, whereas every feature riding the order parameter—the $d_z^{\mathrm{eff}} \propto \langle F_x \rangle$ family *and* the $(E_{12}, q_{\mathrm{AFM}})$ transfer, whose static vertex contracts G_z—must vanish there; the geometry-B same-polarised map must be led by the $2E_{12}$ harmonic essentially alone, any rectified $(E_{12}, 0)$ line falsifying the exclusion of population readout; and adiabatically long pulses collapse the geometry-A cross peak. Propagation adds three fingerprints of its own: a lineout across $\omega_t = 1.1$–1.3 at $\omega_\tau = q_{\mathrm{AFM}}$ must show a self-reversal *doublet* around a dark centre at 1.20; the geometry-B cross channel must carry a d_z^{eff} doublet at $\omega_t = 0.45/0.54$ at the ~ 0.17 level; and every wing amplitude scales with sample thickness, so a thin sample revives the E_{13} line centre and the two-quantum (E_{13}, E_{23}) row together. The temperature scan through the reorientation remains the decisive test, since it separates the symmetry-broken from the conversion-generated features and measures d_z^{eff} by the size of the collapse.

VIII. CONCLUSION

Two components of one symmetry-selected static Fe–Tm vertex family, one Fe-sector magnon–magnon coupling, a mirror-parity rule assigning each polarisation to one qutrit block—so that the drive's electric and magnetic quadratures write different qutrit coordinates, with their ratio w_{E1} measured by a single cross peak—and a hyperpolarisability allowed by the absence of inversion at the rare-earth site—completed by a single reciprocal propagation filter, the self-absorption of the electric-dipole light at the strong CEF lines—account for the measured 2DCS atlas of $TmFeO_3$, computed from measured inputs with no field-assisted coupling and no free lineshape parameters. The two residuals that remain are named, characterised, and funnel jointly into one measurable object: the sample's polarisation-resolved optical depths at the CEF lines. Every peak has been given a pathway, every structural absence a selection rule, and every assignment a falsifiable signature. The program—bound the vertex space by symmetry, assign every feature and every absence, and let the atlas overdetermine the model—applies to any magnet whose 2D map spans distinct subsystems, and turns 2DCS from a spectroscopy of modes into a spectroscopy of Hamiltonian terms.

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Derivations, numerical protocol and the complete falsification ledger are in the Supplemental Material; the long-form note and all run recipes accompany the data.

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